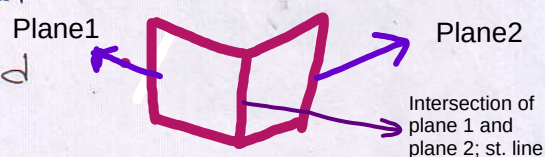


STRAIGHT LINE

General Equation of Straight line.

Every equation of the first degree represents a plane and a plane cuts another plane in a straight line. Therefore in general a line is obtained by the intersection of two planes and as such the two linear equations in x, y, z taken together represent that line.

Let $a_1x + b_1y + c_1z + d_1 = 0$ and $a_2x + b_2y + c_2z + d_2 = 0$



be two planes. Then if (x, y, z) be a point on their line of intersection, it lies on both the planes.

Now if we take two eqs of planes of the 1st degree in x, y and z , together i.e.

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2 \quad \text{--- (1)}$$

then any point which satisfies these (two) eqs individually will satisfy both of them as well. Hence these two eqs taken together represent the line of intersection of the two given planes and we know that two planes intersect in a straight line.

then the eqⁿ (1) represents the eqⁿ of a straight line which is the line of intersection of the two planes.

Thus a straight line in space is represented by the two eqs of 1st degree in x, y, z .

Parametric equations of the line

Let the straight line L passes through the point $A(x_1, y_1, z_1)$ having direction cosines l, m, n respectively. Let $P(x, y, z)$ be any point on the line L and $AP = r$, then L is the locus of P .

[4]

Consider the line L through the point $P = (x_0, y_0, z_0)$ that is parallel to the vector $\vec{V} = \langle a, b, c \rangle$.

The line L consists of all points $Q = (x, y, z)$ for which $\vec{PQ}$ is parallel to $\vec{V}$

Now $\vec{PQ} = \langle x - x_0, y - y_0, z - z_0 \rangle$

Since $\vec{PQ}$ is parallel to $\vec{V} = \langle a, b, c \rangle$

$$\vec{PQ} = t \vec{V}, \quad t \text{ is a scalar}$$

$$\Rightarrow \langle x - x_0, y - y_0, z - z_0 \rangle = t \langle a, b, c \rangle$$

$$\Rightarrow \langle x, y, z \rangle - \langle x_0, y_0, z_0 \rangle = t \langle a, b, c \rangle \Rightarrow \langle x, y, z \rangle = \langle x_0, y_0, z_0 \rangle + t \langle a, b, c \rangle$$

$$\left[\begin{array}{l} \therefore x - x_0 = at \\ y - y_0 = bt \\ z - z_0 = ct \end{array} \right]$$

$$\Rightarrow \vec{r} = \vec{r}_0 + t \vec{V}, \quad \text{where } \vec{r} = \langle x, y, z \rangle$$

$$\vec{r}_0 = \langle x_0, y_0, z_0 \rangle$$

Again $\langle x, y, z \rangle = \langle x_0 + ta, y_0 + tb, z_0 + tc \rangle$

$$\Rightarrow x = x_0 + ta$$

$$y = y_0 + tb$$

$$z = z_0 + tc$$

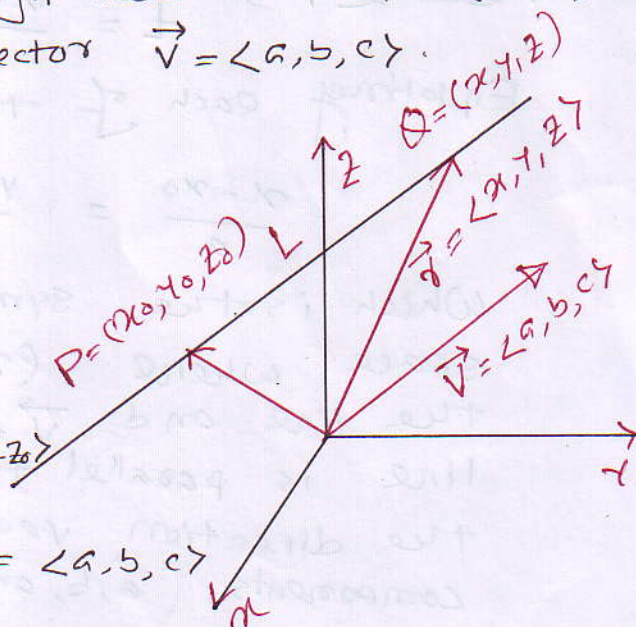
which is called the parametric eqⁿ of a line.

and the vector $\vec{V} = \langle a, b, c \rangle$ is called the direction vector for the line L and its components a, b , and c are called the direction numbers.

Symmetric Eqⁿ of a Line in 3D Space

Assuming $a \neq 0, b \neq 0, c \neq 0$, if we take each parametric eqⁿ and solve for the variable t , we obtain

PTO



the eqns $t = \frac{x-x_0}{a}$, $t = \frac{y-y_0}{b}$, $t = \frac{z-z_0}{c}$

Equating each of these eqns gives

$$\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$$

Which is the symmetric eqns of a line L in 3D space where (x_0, y_0, z_0) is a point passing through the line and $\vec{v} = \langle a, b, c \rangle$ is a vector that the line is parallel to the vector $\vec{v} = \langle a, b, c \rangle$ is called the direction vector for the line L and its components a, b , and c are called the direction numbers.

Note: To write the eqn of a line in 3D space, we need a point on the line and a parallel vector to the line.

Ex: Find the vector, parametric, and symmetric equations for the line through the point $(1, 0, 3)$ and parallel to the vector $2\hat{i} - 4\hat{j} + 5\hat{k}$.

Solⁿ: The given point is $(1, 0, 3)$ and vector

$$\vec{v} = 2\hat{i} - 4\hat{j} + 5\hat{k} = \langle 2, -4, 5 \rangle$$

The vector eqn of the line is

$$\vec{r} = \vec{r}_0 + t\vec{v}$$

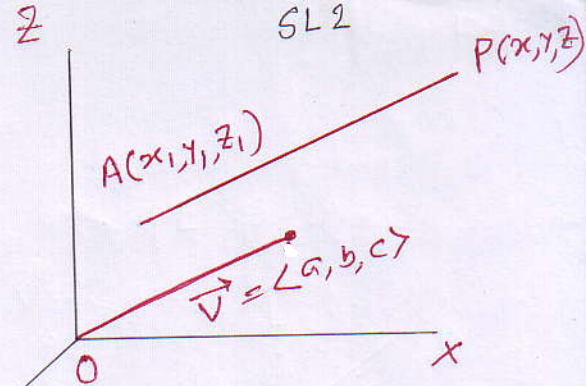
$$\Rightarrow \langle x, y, z \rangle = \langle 1, 0, 3 \rangle + t\langle 2, -4, 5 \rangle$$

parametric eqn is $x = 1 + 2t$, $y = 0 - 4t$, $z = 3 + 5t$

$$\text{Symmetric eqn is } \frac{x-1}{2} = \frac{y-0}{-4} = \frac{z-3}{5}$$

$$\Rightarrow \frac{x-1}{2} = -\frac{y}{4} = \frac{z-3}{5}$$

Taking projections of AP on the coordinate axes (i.e. axes of x, y, z) respectively, we get



$$\left. \begin{aligned} x - x_1 &= r \cos \alpha = r l \\ y - y_1 &= r \cos \beta = r m \\ z - z_1 &= r \cos \gamma = r n \end{aligned} \right\} \text{--- (1)}$$

So for all points (x, y, z) on the given line, we have

$$\left. \begin{aligned} x &= x_1 + r l \\ y &= y_1 + r m \\ z &= z_1 + r n \end{aligned} \right\} \text{--- (2)}$$

which are called the parametric eqⁿs (Freedom eqⁿs) of the line in terms of the parameter r .
The parameter r is the algebraic distance of any point on the line from the point $A(x_1, y_1, z_1)$.
If a, b, c be the direction ratios of the line L , then the above eqⁿs (2) can be written as

$$\left. \begin{aligned} x &= x_1 + at \\ y &= y_1 + bt \\ z &= z_1 + ct \end{aligned} \right\} \text{--- (3)}$$

$$\text{where } t = \frac{r}{\sqrt{a^2 + b^2 + c^2}} = \frac{r}{\sqrt{\sum a^2}}$$

Symmetrical form of the eqⁿs of the line
Symmetrical form in terms of direction cosines

Find the equation of a line whose direction cosines are (l, m, n) and which passes through the point (x_1, y_1, z_1) .

Let P be any point of the line such that $AP = r$

If LM be the projection of AP on the x -axis, then we have

$LM = x - x_1$. Also we have $LM = r l$

$$\therefore x - x_1 = r l$$

$$\Rightarrow \frac{x - x_1}{l} = r.$$

Similarly, considering the projections of AP on the y and z axes respectively we get

$$\frac{y - y_1}{m} = r$$

and $\frac{z - z_1}{n} = r$

$$\text{Thus } \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r \quad \text{--- (1)}$$

which is the symmetrical form of the eqⁿs of a straight line.

▣ Symmetrical form in terms of direction ratios

If a, b, c are the direction ratios of the line then the actual direction cosines of the line are

$$l = \frac{a}{\sqrt{\sum a^2}}, \quad m = \frac{b}{\sqrt{\sum a^2}} \quad \text{and} \quad n = \frac{c}{\sqrt{\sum a^2}}.$$

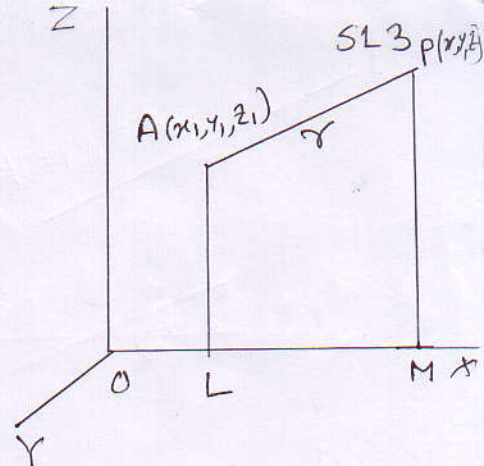
$$\text{Thus the eqⁿs of the line } \frac{x - x_1}{l} = \frac{y - y_1}{m} = \frac{z - z_1}{n} = r,$$

take the form -

$$\frac{x - x_1}{\frac{a}{\sqrt{\sum a^2}}} = \frac{y - y_1}{\frac{b}{\sqrt{\sum a^2}}} = \frac{z - z_1}{\frac{c}{\sqrt{\sum a^2}}} = r$$

$$\Rightarrow \frac{x - x_1}{a} = \frac{y - y_1}{b} = \frac{z - z_1}{c} = R \text{ (say)} \quad \text{--- (2)}$$

Any point on (2) is $(aR + x_1, bR + y_1, cR + z_1)$; but R does not represent the actual length of any point on the line from (x_1, y_1, z_1) since $a^2 + b^2 + c^2 \neq 1$. Here R is a parameter.



□

If the line $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = R$ is perpendicular to one of the co-ordinate axes, the eqⁿ takes one of the following:

$$x=x_1, \frac{y-y_1}{b} = \frac{z-z_1}{c} \text{ (perpendicular to } x\text{-axis)}$$

$$y=y_1, \frac{x-x_1}{a} = \frac{z-z_1}{c} \text{ (" " } y \text{ ")}$$

$$z=z_1, \frac{x-x_1}{a} = \frac{y-y_1}{b} \text{ (" " } z \text{ ")}$$

□ If the line $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c} = R$ is perpendicular to two axes two eq^s are sufficient to determine the line

$$x=x_1, y=y_1 \text{ (} \perp \text{ to } x \text{ and } y \text{ axis)}$$

$$x=x_1, z=z_1 \text{ (} \perp \text{ " } x \text{ " } z \text{ ")}$$

$$y=y_1, z=z_1 \text{ (} \perp \text{ " } y \text{ " } z \text{ ")}$$

□ Line Through two given points

If P and Q are $(x_1, y_1, z_1), (x_2, y_2, z_2)$, the direction ratios of PQ are proportional to $x_2-x_1, y_2-y_1, z_2-z_1$ and therefore the eqⁿ of line PQ are $\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$

Ex. find the parametric and symmetric eq^s of the line through the points $(1, 2, 0)$ and $(-5, 4, 2)$

Solⁿ: - To find the eqⁿ of a line in 3D space, we must have at least one point on the line and a parallel vector. We already have two points on the line so we have at least one point. To find a parallel vector, we can simply just use the vector that passes between two given points, which will also be on this line. That is, if we assign the points $\vec{P} = (1, 2, 0)$ and $\vec{Q} = (-5, 4, 2)$, then the parallel vector $\vec{v}$ is given by $\vec{v} = \vec{PQ} = \langle -5-1, 4-2, 2-0 \rangle = \langle -6, 2, 2 \rangle$

We know the parametric eqⁿ of a line are given by

$$x = x_0 + at, \quad y = y_0 + bt, \quad z = z_0 + ct$$

We can use either the point P or Q our point on the line (x_0, y_0, z_0) . We choose the point P and assign $(x_0, y_0, z_0) = (1, 2, 0)$.

So the parametric eqⁿ are

$$x = 1 + (-6)t, \quad y = 2 + 2t, \quad z = 0 + 2t$$

$$\Rightarrow x = 1 - 6t, \quad y = 2 + 2t, \quad z = 2t$$

Let $P = (x_1, y_1, z_1) = (1, 2, 0)$ and

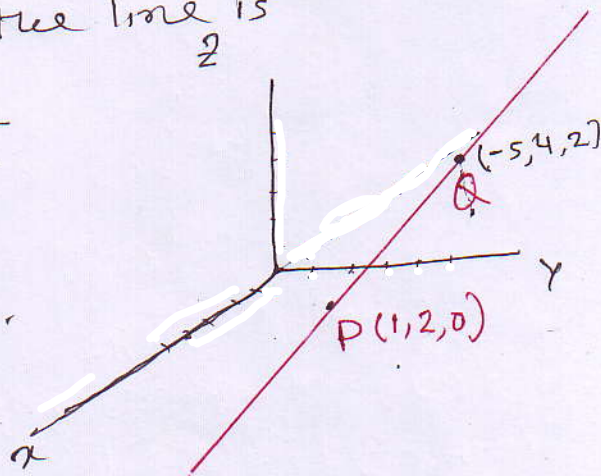
$$Q = (x_2, y_2, z_2) = (-5, 4, 2)$$

So the symmetric eqⁿ of the line is

$$\frac{x - x_1}{x_1 - x_2} = \frac{y - y_1}{y_1 - y_2} = \frac{z - z_1}{z_1 - z_2}$$

$$\Rightarrow \frac{x - 1}{1 + 5} = \frac{y - 2}{2 - 4} = \frac{z - 0}{0 - 2}$$

$$\Rightarrow \frac{x - 1}{6} = \frac{y - 2}{-2} = \frac{z}{-2}$$



It is important to note that the presentation of lines in 3D space are not unique. For instance, in the above example, if we had used the point $Q = (-5, 4, 2)$ to represent the eqⁿ of the line with the parallel vector $\vec{v} = \langle -6, 2, 2 \rangle$, the parametric eqⁿ becomes

$$x = -5 - 6t, \quad y = 4 + 2t, \quad z = 2 + 2t$$

Direction ratios from eqⁿ of a straight line

Let the eqⁿs of the straight line be

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2 \quad \text{--- (1)}$$

Then the eqⁿ of a line which is parallel to (1) and passes through the origin are

$$a_1x + b_1y + c_1z = 0 = a_2x + b_2y + c_2z$$

Now solving eqⁿs

$$\begin{aligned} a_1x + b_1y + c_1z &= 0 \\ a_2x + b_2y + c_2z &= 0 \end{aligned}$$

We have

$$\frac{x}{b_1c_2 - b_2c_1} = \frac{y}{a_2c_1 - a_1c_2} = \frac{z}{a_1b_2 - a_2b_1}$$

Which show the direction ratios of the line proportional to $b_1c_2 - b_2c_1$, $a_2c_1 - a_1c_2$, $a_1b_2 - a_2b_1$ multiple of one another or

▢ If the direction ratios of two lines are same, then they are parallel.

Ex: Show that lines

$$5x - y - 3z + 12 = 0 = x - 7y + 5z - 6$$

$$2x + 3y - 4z = 0 = 3x - 4y + z - 4$$

are parallel.

Solⁿ: The direction ratio of 1st line is

$$b_1c_2 - b_2c_1, a_2c_1 - a_1c_2, a_1b_2 - a_2b_1$$

or

$$3 - 16, -12 - 2, -8 - 9$$

$$\begin{vmatrix} 2 & 3 & -4 & 2 & 3 \\ 3 & -4 & 1 & 3 & -4 \end{vmatrix}$$

or

$$-13, -14, -17$$

The direction ratio of 2nd line is -

or

$$-5 - 21, -3 - 25, -35 + 1$$

$$\begin{vmatrix} 5 & -1 & -3 & 5 & -1 \\ 1 & -7 & 5 & 1 & -7 \end{vmatrix}$$

or $-26, -28, -34$

Since the direction ratio of the 2nd line is multiple of direction ratio of 1st line, so they are parallel.



Transformation of eqⁿ of straight line from general form to symmetrical form.

Let the eqⁿs of the line in general form be

$$a_1x + b_1y + c_1z + d_1 = 0 = a_2x + b_2y + c_2z + d_2 \quad \text{--- (1)}$$

We know the symmetric eqⁿ of a st. line is $\frac{x-x_0}{a} = \frac{y-y_0}{b} = \frac{z-z_0}{c}$ which is passed through (x_0, y_0, z_0) and the direction ratios are proportional to $\{a, b, c\}$. So to find the symmetric form we have to find a point on the line and direction ratios which are proportional

To find a point on (1), let $z=0$, then (1) $\Rightarrow$

$$a_1x + b_1y + d_1 = 0 \quad \text{--- (2)}$$

$$a_2x + b_2y + d_2 = 0 \quad \text{--- (3)}$$

$$\Rightarrow \frac{x}{b_1d_2 - b_2d_1} = \frac{y}{a_2d_1 - a_1d_2} = \frac{1}{a_1b_2 - a_2b_1}$$

$\therefore$ The required point is $\left(\frac{b_1d_2 - b_2d_1}{a_1b_2 - a_2b_1}, \frac{a_2d_1 - a_1d_2}{a_1b_2 - a_2b_1}, 0 \right)$

In case $a_1b_2 - a_2b_1 = 0$, then instead of $z=0$ we may put $x=0$ or $y=0$.

Thus, in the symmetrical form, the eqⁿ of the given line whose direction ratios are proportional to $b_1c_2 - b_2c_1$, $a_2c_1 - a_1c_2$, $a_1b_2 - a_2b_1$ and which passes through a point $(x_1, y_1, 0)$ are

$$\frac{x-x_1}{b_1c_2 - b_2c_1} = \frac{y-y_1}{a_2c_1 - a_1c_2} = \frac{z-0}{a_1b_2 - a_2b_1}$$

Ex: Find the eqⁿ of the line $3x - 4y + 2z + 5 = 0 = 2x + 3y - 5z - 8$ in the symmetrical form.

Solⁿ: Let $z = 0$, then $3x - 4y + 5 = 0$ — (1)
 $2x + 3y - 8 = 0$ — (2)

$$\therefore \frac{x}{32-15} = \frac{y}{10+24} = \frac{1}{9+8}$$

$$\Rightarrow x = \frac{17}{17} = 1$$

$$y = \frac{34}{17} = 2$$

$$\therefore (x, y, z) = (1, 2, 0)$$

Now for direction ratios

$$3 \quad -4 \quad 2 \quad 3 \quad -4$$

$$2 \quad 3 \quad -5 \quad 2 \quad 3$$

$$\Rightarrow 20 - 6, \quad 4 + 15, \quad 9 + 8$$

$$\Rightarrow 14, \quad 19, \quad 17$$

So the direction ratio of the given line is proportional to 14, 19, 17.

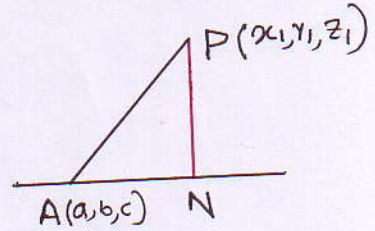
So the symmetrical form of the given line which passes through $(1, 2, 0)$ and whose direction ratios are proportional to 14, 19, 17 is

$$\frac{x-1}{14} = \frac{y-2}{19} = \frac{z-0}{17}$$

Distance of a point from a line

Let the length of the perpendicular distance from a point $P(x_1, y_1, z_1)$ to the line $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-c}{n}$ be P

Let A be the point (a, b, c) on the given line and N the foot of the perpendicular from P on it.



$$\text{We have } PN^2 = AP^2 - AN^2$$

$$\text{But } AP^2 = (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2$$

and $AN = \text{projection of } AP \text{ on the given line}$
 $= l(x_1 - a) + m(y_1 - b) + n(z_1 - c)$ where l, m, n are the actual direction cosine of AP .

$$\begin{aligned} \therefore PN^2 &= (x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2 - \{l(x_1 - a) + m(y_1 - b) + n(z_1 - c)\}^2 \\ &= \{(x_1 - a)^2 + (y_1 - b)^2 + (z_1 - c)^2\} (l^2 + m^2 + n^2) \\ &\quad - l^2(x_1 - a)^2 - m^2(y_1 - b)^2 - n^2(z_1 - c)^2 - 2lm(y_1 - a)(y_1 - b) \\ &\quad - 2mn(y_1 - b)(z_1 - c) - 2nl(z_1 - c)(x_1 - a) \\ &= \cancel{(x_1 - a)^2 l^2} + \cancel{(x_1 - a)^2 m^2} + \cancel{(x_1 - a)^2 n^2} + \cancel{(y_1 - b)^2 l^2} + \cancel{(y_1 - b)^2 m^2} + \cancel{(y_1 - b)^2 n^2} \\ &\quad + \cancel{(z_1 - c)^2 l^2} + \cancel{(z_1 - c)^2 m^2} + \cancel{(z_1 - c)^2 n^2} - \cancel{(x_1 - a)^2 l^2} - \cancel{(y_1 - b)^2 m^2} - \cancel{(z_1 - c)^2 n^2} \\ &\quad - 2lm(x_1 - a)(y_1 - b) - 2mn(y_1 - b)(z_1 - c) - 2nl(z_1 - c)(x_1 - a) \\ &= \{(x_1 - a)m - (y_1 - b)l\}^2 + \{(y_1 - b)n - (z_1 - c)m\}^2 \\ &\quad + \{(z_1 - c)l - (x_1 - a)n\}^2 \\ &= \left| \begin{array}{cc} x_1 - a & y_1 - b \\ l & m \end{array} \right|^2 + \left| \begin{array}{cc} y_1 - b & z_1 - c \\ m & n \end{array} \right|^2 + \left| \begin{array}{cc} z_1 - c & x_1 - a \\ n & l \end{array} \right|^2 \end{aligned}$$

Ex: Find the distance between point $P(0, 2, 3)$ and the line $\frac{x-3}{2} = \frac{y-1}{1} = \frac{z+1}{2}$.

Soln: - If P is the distance then

$$\begin{aligned}
 P^2 &= \begin{vmatrix} x_1 - a & y_1 - b \\ l & m \end{vmatrix}^2 + \begin{vmatrix} y_1 - b & z_1 - c \\ m & n \end{vmatrix}^2 + \begin{vmatrix} z_1 - c & x_1 - a \\ n & l \end{vmatrix}^2 \\
 &= \begin{vmatrix} 0-3 & 2-1 \\ 2 & 1 \end{vmatrix}^2 + \begin{vmatrix} 2-1 & 3-(-1) \\ 1 & 2 \end{vmatrix}^2 + \begin{vmatrix} 3-(-1) & 0-3 \\ 2 & 2 \end{vmatrix}^2 \\
 &= (-3-2)^2 + (2-4)^2 + (8+6)^2 \\
 &= 25 + 4 + 196 \\
 &= 225
 \end{aligned}$$

$$\therefore P = 15$$

Ex: Find the distance of the point $(-1, 2, 5)$ from the str. line $\frac{x-3}{2} = \frac{y-4}{-\frac{3}{7}} = \frac{z-5}{\frac{6}{7}}$.

$$\begin{aligned}
 \text{Soln. } P^2 &= \begin{vmatrix} -1-3 & 2-4 \\ \frac{2}{7} & -\frac{3}{7} \end{vmatrix}^2 + \begin{vmatrix} 2-4 & 5-5 \\ -\frac{3}{7} & \frac{6}{7} \end{vmatrix}^2 + \begin{vmatrix} 5-5 & -1-3 \\ \frac{6}{7} & \frac{2}{7} \end{vmatrix}^2 \\
 &= \left(\frac{12}{7} + \frac{4}{7}\right)^2 + \left(-\frac{12}{7}\right)^2 + \left(\frac{24}{7}\right)^2 \\
 &= \frac{1}{49} (256 + 144 + 576) \\
 &= \frac{976}{49} = \frac{16 \times 61}{49} \\
 \therefore P &= \frac{4\sqrt{61}}{7}
 \end{aligned}$$

Ex: Find the distance of the point $(-1, -5, -10)$ from the point of intersection of the line

$$\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} \text{ and the plane } x-y+z=5.$$

Solⁿ: Let $\frac{x-2}{3} = \frac{y+1}{4} = \frac{z-2}{12} = r$

$$\therefore x = 3r+2, y = 4r-1, z = 12r+2$$

The given line meets the plane $x-y+z=5$

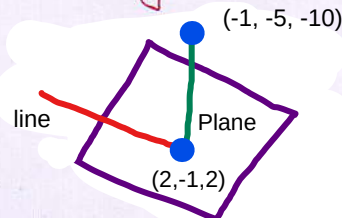
$$\therefore 3r+2 - 4r+1 + 12r+2 = 5$$

$$\Rightarrow r = 0$$

$$\text{so } (x, y, z) = (2, -1, 2)$$

Therefore distance between $(-1, -5, -10)$ and $(2, -1, 2)$ is

$$\begin{aligned} & \sqrt{(-1-2)^2 + (-5+1)^2 + (-10-2)^2} \\ &= \sqrt{9+16+144} \\ &= \sqrt{169} = 13 \end{aligned}$$



Ex: Find the shortest distance between the lines

$$\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1}, \quad \frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4}$$

Find also the eqⁿ and the points in which it meets the given lines.

Solⁿ: Let $\frac{x-3}{3} = \frac{y-8}{-1} = \frac{z-3}{1} = r_1 \Rightarrow x = 3r_1+3$
 $y = -r_1+8$
 $z = r_1+3$

and $\frac{x+3}{-3} = \frac{y+7}{2} = \frac{z-6}{4} = r_2 \Rightarrow x = -3r_2-3$
 $y = 2r_2-7$
 $z = 4r_2+6$

Let $L(3r_1+3, -r_1+8, r_1+3)$; $M(-3r_2-3, 2r_2-7, 4r_2+6)$
 be any two points L, and M on the given lines

The direction ratios of the LM are

$$3r_1+3r_2+6, -r_1-2r_2+15, r_1-4r_2-3$$

If the Line LM is perpendicular to both the given lines, then we have

$$3(3x_1 + 3x_2 + 6) - 1(-x_1 - 2x_2 + 15) + (x_1 - 4x_2 - 3) = 0 \Rightarrow 11x_1 + 7x_2 = 0$$

$$\text{and } -3(3x_1 + 3x_2 + 6) + 2(-x_1 - 2x_2 + 15) + 4(x_1 - 4x_2 - 3) = 0 \Rightarrow 7x_1 + 20x_2 = 0$$

The above two eqns give $x_1 = x_2 = 0$.

$$\therefore L \equiv (3, 8, 3)$$

$$M = (-3, -7, 6)$$

and the direction ratios of the LM are 6, 15, -3

The eqn of the line LM which is the S.D is

$$\frac{x-3}{3+3} = \frac{y-8}{8+7} = \frac{z-3}{3-6}$$

$$\text{or } \frac{x-3}{6} = \frac{y-8}{15} = \frac{z-3}{-3}$$

$$\text{or } \frac{x-3}{2} = \frac{y-8}{5} = \frac{z-3}{-1}$$

The length of S.D between L and M is

$$\sqrt{(3+3)^2 + (8+7)^2 + (3-6)^2}$$

$$= \sqrt{36 + 225 + 9}$$

$$= \sqrt{270}$$

$$= 3\sqrt{30}$$

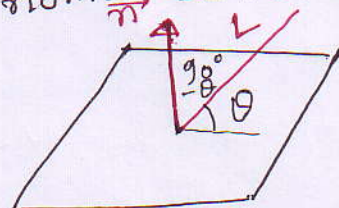
Angle between a Line and a Plane

Let the line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ — (I)

— (II)

and " plane $ax+by+cz+d=0$

Angle between a line and a plane is the complement of the angle between the line and normal $\vec{n}$ to the plane



Let θ be the angle between the line and the plane. Then the angle between the line and the normal to the plane will be $90^\circ - \theta$

Since the direction cosines of the normal to the given plane and of the given line are proportional to a, b, c and l, m, n respectively, we have

$$\cos(90^\circ - \theta) = \sin \theta = \frac{al + bm + cn}{\sqrt{a^2 + b^2 + c^2} \sqrt{l^2 + m^2 + n^2}}$$

where θ is the angle between the plane and line and $90^\circ - \theta$ is the angle between the line and the normal.

Condition of Parallelism

The st. line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ and the plane

$ax+by+cz+d=0$ will be parallel if $\theta=0$ i.e.

$$al + bm + cn = 0$$

which is also evident from the fact that a line be parallel to a plane if it is $\perp$ to the normal to plane.

Condition of Perpendicularity of a line and a plane

The line $\frac{x-x_1}{l} = \frac{y-y_1}{m} = \frac{z-z_1}{n}$ and the plane $ax+by+cz+d=0$ will be $\perp$ if the normal to the plane and the line are parallel i.e. $\frac{l}{a} = \frac{m}{b} = \frac{n}{c}$.

Ex: Find the distance of the point $(1, -2, 3)$ from the plane $x - y + z - 5 = 0$ measured parallel to the line

$$\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$$

Soln: Eqⁿ of the line through $A(1, -2, 3)$ parallel to the line $\frac{x}{2} = \frac{y}{3} = \frac{z}{-6}$ is

$$\frac{x-1}{2} = \frac{y+2}{3} = \frac{z-3}{-6} = r \text{ (say)} \quad \text{--- (1)}$$

Then any point on the line (1) is

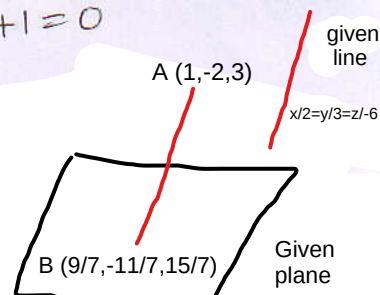
$$(2r+1, 3r-2, -6r+3) \quad \text{--- (2)}$$

If this point lies on the plane $x - y + z - 5 = 0$ then

$$2r+1 - 3r+2 - 6r+3 - 5 = 0 \Rightarrow -7r+1=0$$

$$\Rightarrow r = 1/7$$

$$\therefore (2) \Rightarrow \left(\frac{2}{7} + 1, \frac{3}{7} - 2, -\frac{6}{7} + 3 \right) \\ = \left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7} \right)$$



which is the point of intersection of the line⁽¹⁾ and the plane $x - y + z - 5 = 0$

Now distance between $(1, -2, 3)$ and $\left(\frac{9}{7}, -\frac{11}{7}, \frac{15}{7}\right)$ is

$$P = \sqrt{\left(1 - \frac{9}{7}\right)^2 + \left(-2 + \frac{11}{7}\right)^2 + \left(3 - \frac{15}{7}\right)^2} = 1.$$

Ex: Find the eqⁿ of the plane through the line $3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4$ and parallel to the line $x = 2y = 3z$

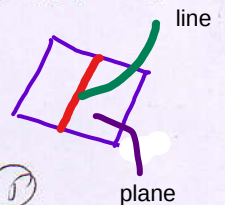
Soln: The given eq^s of the lines are -

$$3x - 4y + 5z - 10 = 0 = 2x + 2y - 3z - 4 \quad \text{--- (1)}$$

$$\text{and } x = 2y = 3z \Rightarrow \frac{x}{6} = \frac{y}{3} = \frac{z}{2} \quad \text{--- (2)}$$

The eqⁿ of any plane through the line (1) is

$$3x - 4y + 5z - 10 + K(2x + 2y - 3z - 4) = 0$$



$$\Rightarrow (3+2K)x + (-4+2K)y + (5-3K)z + (-10-4K) = 0 \quad \text{--- (3)}$$

If this plane (3) is parallel to the line (2), then we have

$$(3+2K) \cdot 6 + (-4+2K) \cdot 3 + (5-3K) \cdot 2 = 0$$

$$\Rightarrow 12K + 16 = 0 \Rightarrow K = -\frac{4}{3}$$

Putting $K = -\frac{4}{3}$ in eqn (3) we have

$$\left(3 - \frac{8}{3}\right)x + \left(-4 - \frac{8}{3}\right)y + (5+4)z + \left(-10 + \frac{16}{3}\right) = 0$$

$$\Rightarrow x - 20y + 27z - 14 = 0$$

Ex: Find the eqn of the line through the point (1,2,3) parallel to the line $x-y+2z-5=0 = 3x+y+z-6$

Solⁿ:

Let l, m, n be the direction cosines of the line. Then since this line is parallel to both of the planes, so it is perpendicular to the normal of these planes. Therefore, we have

$$l - m + 2n = 0 \quad \text{--- (1)}$$

$$3l + m + n = 0 \quad \text{--- (2)}$$

$$\Rightarrow \frac{l}{-1-2} = \frac{m}{6-1} = \frac{n}{1+3}$$

$$\Rightarrow \frac{l}{-3} = \frac{m}{5} = \frac{n}{4}$$

$\therefore$ The direction ratios of the line are (-3, 5, 4)
Hence the required eqn of the line through (1,2,3) is

$$\frac{x-1}{-3} = \frac{y-2}{5} = \frac{z-3}{4}$$

Ex: Find the eqⁿs of the line which passes through the point $(3, -1, 11)$ and is perpendicular to the line

$\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$. Also obtain the foot of the perpendicular.

Solⁿ: The eqⁿs of the given line are $\frac{x-0}{2} = \frac{y-2}{3} = \frac{z-3}{4} = r(\text{say})$ — (1)

The co-ordinates of any point on the given line (1) are $(2r, 3r+2, 4r+3)$ — (2)

This will be the required foot of the perpendicular if the line joining it to the point $(3, -1, 11)$ be perpendicular to the given line.

Now the direction ratios of the perpendicular line joining $(3, -1, 11)$ and $(2r, 3r+2, 4r+3)$ are $2r-3, 3r+3, 4r-8$

Since the required line is $\perp$ to $\frac{x}{2} = \frac{y-2}{3} = \frac{z-3}{4}$, so

$$2(2r-3) + 3(3r+3) + 4(4r-8) = 0$$

$$\Rightarrow 29r - 29 = 0 \Rightarrow r = 1$$

Putting $r = 1$ in (2) we get the required foot is

$$(2, 5, 7)$$

And the direction ratios are $2-3, 3+3, 4-8$ i.e. $-1, 6, -4$

Hence the perpendicular line passes through the point $(3, -1, 11)$ and its direction ratios are $(-1, 6, -4)$.

Thus the required eqⁿs of the perpendicular line are

$$\frac{x-3}{-1} = \frac{y+1}{6} = \frac{z-11}{-4}$$